User Analysis

ineffectiveness

Limitations of AI Tools
Basic operations are not AI strengths.
Modern AI models (diffusion, Transformer) focus on semantic understanding, style and content generation

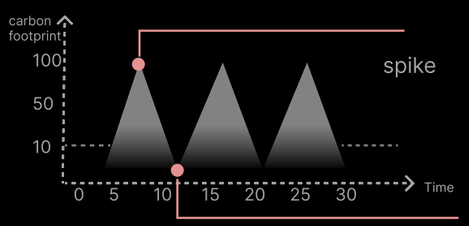
User Cognition and Usage Misconceptions (Primary Cause)
Treating AI as an "all-purpose tool" and ignoring the professionalism of basic operations.

prevalence

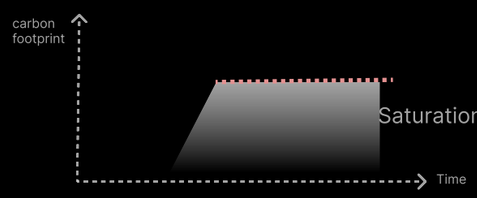
Inefficiency and failure of AI in basic geometric operations are widespread, occurring in approximately 70%–80% of ordinary users and 30%–50% of professional users.

The pattern modelling
Repeated

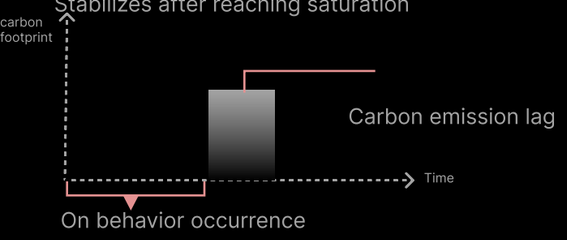
The user performs zoom/rotate operations at a high frequency within a short time, in an uninterrupted loop, far exceeding the number of actions needed for normal viewing or editing.



Click instant

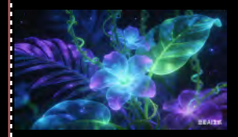


Stabilizes after reaching saturation



On behavior occurrence

Create a sci-fi natural scene with tropical plants that glow with fluorescent light.



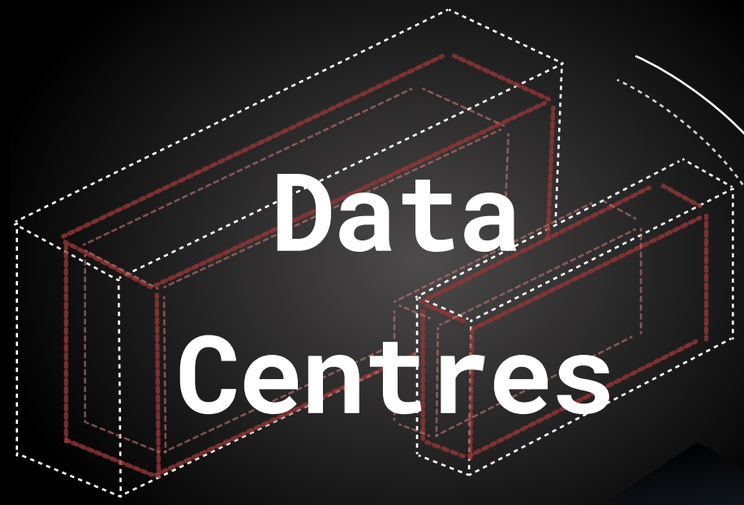
Rotate the image

Upscale the image

Downscale the image

Flip the image vertically

Any transformation to an image's physical form, including size and orientation, creates no new visual information or artistic value. Though users deem it convenient, it is far less efficient than basic operations in image-editing software

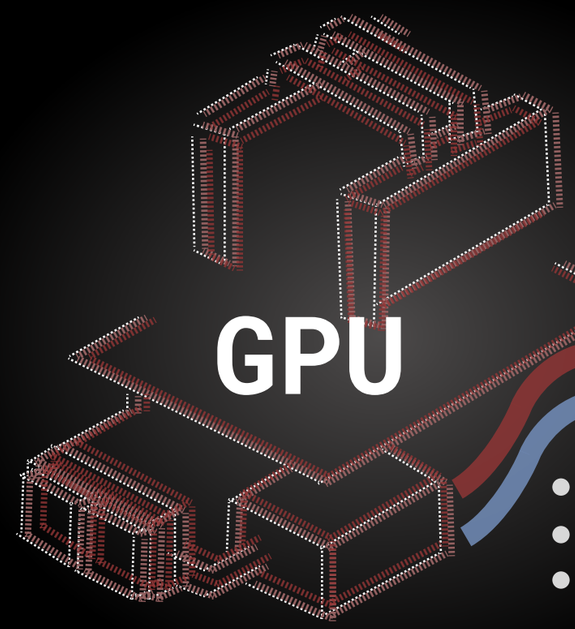


Additional carbon emissions ≈ Emissions from a single image generation

- **Light-weight image repetition: +10–30 g CO_2e**
- **High-quality image repetition: +50–100 g CO_2e**

Main Energy Consumption Components 70%

Servers (especially GPUs) generate heat during operation, requiring water cooling or evaporative cooling to dissipate the heat, during which water is evaporated or discharged.
Regions with higher temperatures and more densely packed computing clusters consume more cooling water.



Heat Generation

CO_2

Cooling Generation

This device only requires a computer cooling fan (bare unit, no complex casing required).
Core Logic

- Behavior Recognition: The AI detects meaningless computational tasks.
- Physical Feedback: The fan starts to spin.

Objective Display Method

- Rotation Speed Corresponding to Load: The rotation speed of the fan directly corresponds to the AI's carbon dioxide emissions.
  - When you perform a zoom operation → the fan gives a brief "hum" and spins once.
  - When you generate content frequently and repeatedly → the fan spins rapidly at high speed.
  - When you stop operating → the fan slows down and gradually stops.

Cooling Distribution

Rotate
Upscale
Downscale
Flip

The AI can detect

Coolant flows over hot chips

H_2O

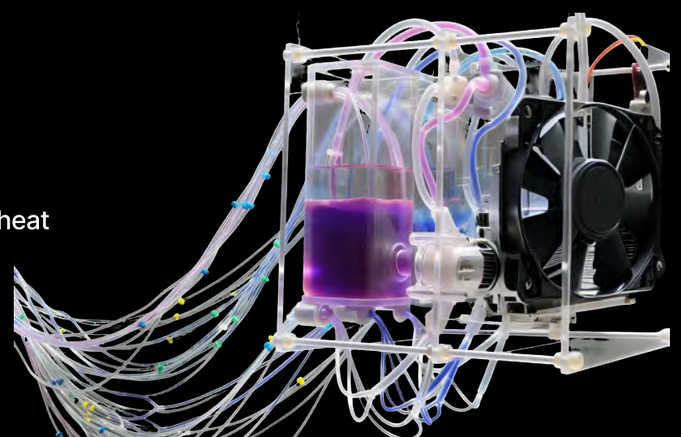
This device only requires a small hydraulic actuator.
Core Logic

- Core Logic
- Physical Feedback: Use the wind energy from the previous step to drive the hydraulic actuator to start operating.

Objective Display Method

- Pressure Corresponding to Load: The operating pressure of the hydraulic actuator directly corresponds to the water required for cooling during AI generation.

Temperature decreases

Cooling fan exhausts heat

- Carbon Dioxide Emissions
- Consumed Water Resources
- Cooling Circulation System